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Detecting Hallucinations in Large Language Models Using Machine Learning Classifiers: A Comparative Study

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Abstract—This study is concerned with the comprehensive evaluation of machine learning models intended for the detection of hallucination in Large Language Model output. The preprocessing of the prompt and answers using TF-IDF vectorization has been done as a preliminary step before developing a hallucination prediction model. Afterwards, the generated dataset was then divided into training and testing, with percentages equal to 70% and 30%, respectively. The selected machine learning models were trained with the dataset created. In this research, eight various classifiers are implemented and compared. To evaluate the results of our approach, multiple metrics were taken into consideration: accuracy, precision, recall, and F1-score. While the training results showed exceptional performance with accuracies exceeding 99% in several models, the test results indicated huge challenges in generalization. Random Forest had the highest testing accuracy of 94.77% at 65% precision but with a very low recall of 4.85%. Besides, cross validation is used with random forest and decision tree, the results showed that the cross-validation is better than the test and train splitting model. The study shows the significant trade-offs between precision and recall in hallucination detection, which is especially problematic while working with imbalanced datasets.

Keywords- TF-IDF, Hallucination, large language model, machine learning models, Cross validation.

I. INTRODUCTION

Recently, Large Language Models have become one of the strongest machinery for natural language processing, matching extremely well across simple text generation to harder reason tasks [1]. Intrinsic to both architectural design and training methodology, the problem of hallucination, in Large Language Models (LLMs), is a broader issue. Unlike any previous AI system, LLMs are trained on the grounds of vast text corpora, where they learn to predict the most probable succeeding tokens given observed patterns, rather than genuinely understanding or reasoning [4]. While useful for many tasks, this approach may lead to the non-factual generation of plausible information. We have recently seen that even advanced models such as GPT-4 and PaLM have hallucination rates of 3% to 27% (or more), based on the complexity and domain of the task [5].

A great deal of interest has been shown in developing robust mechanisms to prevent and detect hallucinations, given the prevalence and consequences behind them. For the notion of LLM hallucination, traditional methods like simple fact-checking on Knowledge bases or using low-level confidence scores, do not work as they don't capture the nuanced nature of hallucinations. Such limitations opened the door for more sophisticated machine-learning approaches to be used for hallucination detection and prediction [7].

Although this challenge has arisen, machine learning has emerged as a promising solution already, bringing powerful pattern recognition and anomaly detection tools to the table which we can utilize for identifying hallucinations. However, with advancements in computational capabilities and algorithmic innovation more advanced detection systems have been capable of developing [8]. Hallucination detection has been applied through machine learning to various technological domains: Natural Language Processing (NLP), pattern recognition, and knowledge representation [9].

Its rapid progress has taken this field from fairly elementary statistics methods into sophisticated neural architectures. Today, state-of-the-art systems have taken advantage of the power of deep learning, transformer architectures and knowledge graphs to develop more reliable detection mechanisms [10]. While these advancements represent a real step forward in dealing with the hallucination problem in LLMs, notable work continues to make these techniques better. Precisely where the opportunity to work out effective hallucination detection systems coexist with large amounts of computational power, sophisticated algorithms, and enormous training datasets creates unexampled opportunity. There is a very important frontier in the intersection of LLM hallucinations and machine learning solutions. In addition, the challenge should have a solution to the underlying mechanisms of hallucination and should build efficient, robust and scalable detection systems. Given the growth of LLMs and their potential applications, reliable hallucination detection will become ever more important for their powerful systems to remain trusted and essentially applicable [11].

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Table I: sample of the TF-IDF results for prompt and answer

		Prompt			Answer		
Answer_n	Prompt_n	an	ancient	and	to	steps	Target
113	88	0.038985	0	0.076525	0.06343	0	0
20	343	0.02395	0	0.105778	0.105246	0	0
74	95	0.071987	0	0.105978	0.047367	0	0
1	562	0.14828	0	0.206168	0	0.025132625	0
2	249	0.014514	0	0.042734	0	0.029520238	0
2	31	0.10321	0	0.101297	0	0	0
4	186	0.095254	0	0.140232	0	0	0
34	74	0.048874	0	0.095936	0	0.099406856	0
86	27	0	0	0.096533	0.187336	0	0
1	24	0	0	0.078069	0	0	1

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II. HALLUCINTION DETECTION SYSTEM BASED ON MACHINE LEARNING MODELS

A. Dataset collection

To develop hallucination classifier models, the first step is to select a hallucination dataset that includes a prompt, answer, and hallucination variable (target). The target variable is generated by using heuristic labelling if the answer is a hallucination as discussed in [16]. After that, the preprocessing techniques are used to convert the collected dataset to Term Frequency-Inverse Document Frequency Vectorizer (TF-IDF). The TF-IDF results will be used to develop a detection system for the hallucination cases.

B. Preprocessing the dataset

The dataset compiled has two features which are ‘Answer’ and ‘Prompt’ features and both belong to the textual data type. The text in these columns is preprocessed and then converted into TF-IDF vectors indicating the importance of the words in every document. These vectors are then employed in classification aiming at identifying hallucination cases in the LLM. This means that the model input will be the initial dataset while the outputs are the X and y variables that shall be used to build a prediction model. The process of converting the original dataset into the X matrix (which contains all the features extracted from the 'Answer' and 'Prompt' columns) involves the following steps:

1. Handle missing values in the "Answer" column.
2. Count the number of words in "Prompt" and "Answer" and store them in Prompt_n and Answer_n.
3. Create a TF-IDF Vectorizer for "Answer".
4. Create a TF-IDF Vectorizer for "Prompt".
5. Apply TF-IDF transformation to "Answer".
6. Apply TF-IDF transformation to "Prompt".
7. Combine TF-IDF features and additional columns into a single feature set (X).
8. Extract target labels (y).

After applying the previous steps, the X matrix contains all the TF-IDF features from the "Answer" and "Prompt" columns, the word count of the answers, and the word count of the prompts. The sample results of the preprocessing steps are shown in Table I. The generated matrix is then divided into X for input and y for target.

C. The Developed Hallucination Detection System

To develop hallucination prediction models, various machine learning algorithms are used based on their robustness and insights derived from previous research. These models include XGBoost (XGB), Random Forest (RF), Decision Tree (DT), Gaussian Naive Bayes (GaussianNB), Adaptive Boosting (AdaBoost), Bagging (Bootstrap Aggregating), Light Gradient Boosting Machine (lgb.LGBM), and CatBoost. Before developing prediction models for the hallucination cases, the output of the preprocessing techniques is used. The X, and y data are divided into training (70%) and testing (30%) datasets. The datasets are used to develop prediction models by feeding the training dataset, afterwards the trained Hallucination prediction models are tested using the testing dataset to validate the capability of the proposed model in understanding the unseen data.

The performance of hallucination prediction is evaluated using various metrics, with a confusion matrix serving as a key tool to visualize the relationship between correct and incorrect classifications. The confusion matrix consists of four components: True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN). In this study, selected classification metrics are used to assess the model's performance, including recall, precision, accuracy, and F1 score, as detailed below.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (1)$$

$$precision = \frac{TP}{TP + FP} \quad (2)$$

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Table II: The results of the hallucination prediction using the training dataset

Classifier	Accuracy	Precision	Recall	F1-score	RoC_AUC
XGB	0.9993	0.7171	1	0.8352	0.9993
RandomForest	0.9993	0.9889	0.9984	0.9936	0.9997
DecisionTree	0.9994	0.9889	1	0.9944	0.9999
GaussianNB	0.8454	0.2574	1	0.4094	0.9189
AdaBoost	0.8042	0.1848	0.778	0.2987	0.884
Bagging	0.992	0.9836	0.8642	0.9201	0.999
lgb.LGBM	0.9766	0.6963	1	0.821	0.9993
CatBoost	0.9776	0.705	1	0.827	0.9992

Table III: The results of the hallucination prediction using test dataset

Classifier	Accuracy	Precision	Recall	f1-score	RoC_AUC
XGB	0.9155	0.2586	0.3097	0.2818	0.7712
RandomForest	0.9477	0.65	0.0485	0.0903	0.7594
DecisionTree	0.9023	0.1852	0.2425	0.21	0.5902
GaussianNB	0.7621	0.0983	0.4216	0.1595	0.6118
AdaBoost	0.7893	0.1452	0.6007	0.2338	0.7621
Bagging	0.9425	0.381	0.1194	0.1818	0.7321
lgb.LGBM	0.9105	0.2472	0.3284	0.2821	0.7934
CatBoost	0.9119	0.2463	0.3134	0.2759	0.7964

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F1\ score\ = 2 * \frac{Precision\ x\ Recall}{Precision\ +\ Recall} \quad (4)$$

True Positives (TP) represent the cases accurately identified as hallucinations. True Negatives (TN) are the cases correctly classified as not hallucination. False Positives (FP) occur when cases are wrongly predicted as hallucinations but are not. False Negatives (FN) happen when actual hallucination cases are misclassified as not hallucinations.

III. Results Analysis and Discussion

To develop a hallucination prediction model using different classification models. First, the collected dataset is divided into training and testing datasets. The training dataset is used to train the classification models, after that, the testing dataset is used to validate the capability of the proposed models in detecting the hallucination cases in the LLM. The cutoff of train and test are 70%, and 30%, respectively. The selected datasets are chosen randomly without prior knowledge of the selected data.

The training results are shown in Table II and Figure 1, The results demonstrated that most classifiers performed well based on different performance metrics including precision, recall, F1-score, and RoC_AUC, with accuracies exceeding 99% for XGBoost (XGB), RandomForest, DecisionTree, and Bagging. DecisionTree marginally outperformed others with an accuracy of 99.94%, showcasing its ability to handle the dataset's complexity. For instance, DecisionTree and RandomForest achieved near-perfect scores across all these metrics, indicating excellent predictive capabilities. In contrast, classifiers like GaussianNB and AdaBoost struggled, with accuracies of 84.54% and 80.42%, respectively. Their lower precision (25.74% for GaussianNB and 18.48% for AdaBoost) suggested difficulty in correctly identifying positive samples, although both maintained perfect recall. This imbalance is reflected in their relatively low F1 scores (40.94% for GaussianNB and 29.87% for AdaBoost). These results hinted that some classifiers were performing well, while others needed to be optimized a lot to perform comparably to the top-performing models. The results in Table III and Figure 1 offered a comprehensive evaluation of the eight classifiers for hallucination prediction, measured by the metrics of accuracy, precision, recall, F1-score, and RoC_AUC. Each of these metrics has shed light on different aspects of performance by these classifiers, bringing into view significant trade-offs between them. High accuracy across most models suggests general effectiveness in correctly classifying examples, yet deeper analysis shows discrepancies in handling positive hallucination cases, as reflected in other metrics like precision and recall. Most classifiers give a very high accuracy proportion of correct predictions, with RandomForest topping at 94.77%, closely followed by Bagging at 94.25%. These values suggest a very good general performance but mask weaknesses in minority class predictions, such as hallucination detection, due to

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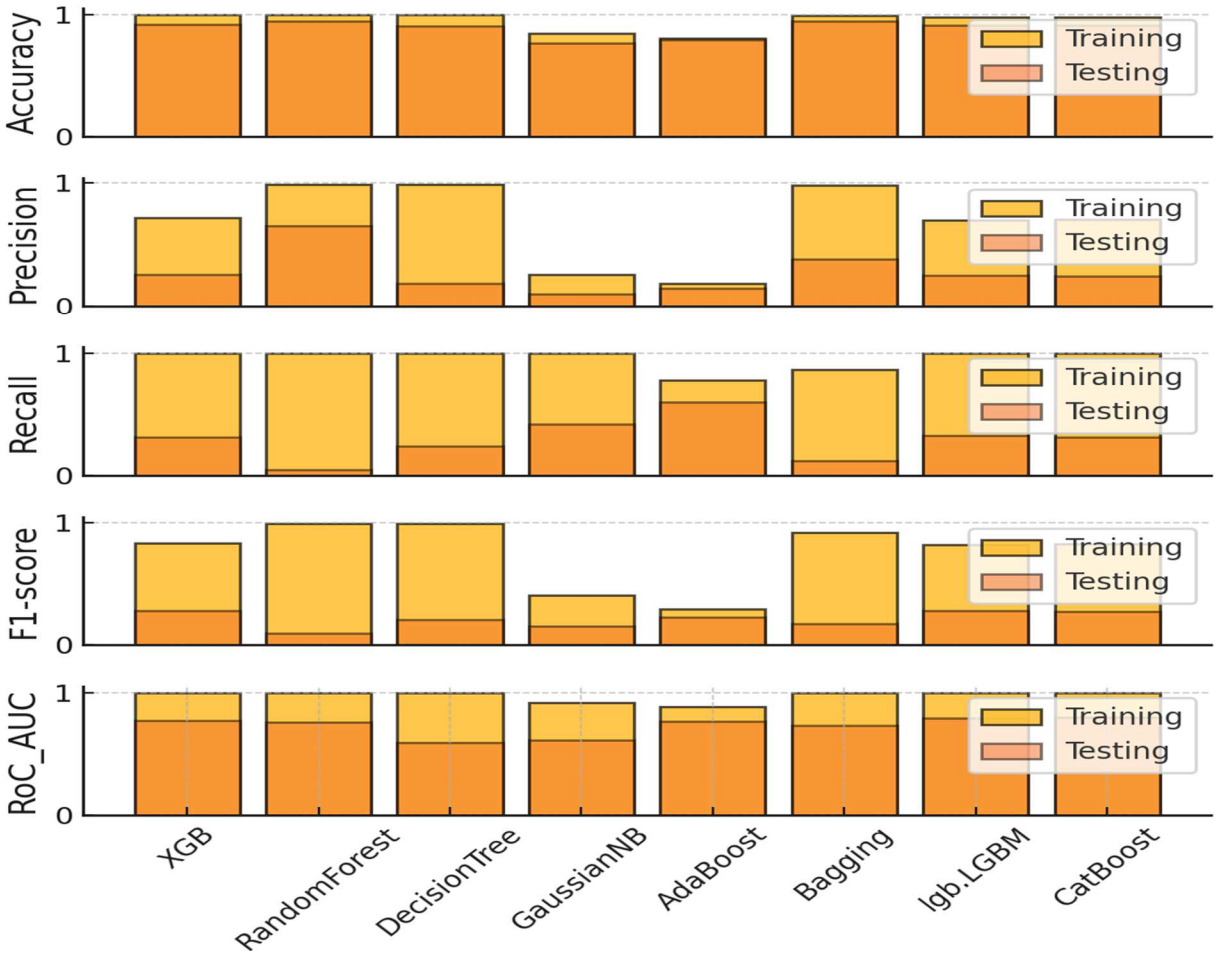


Figure 1: Hallucination detection performance based on machine learning models

potential class imbalance issues in the dataset. High accuracy itself is not sufficient for reliable performance, especially in positive predictions and has wide variations among classifiers. RandomForest presents the highest precision with 65%, which is indicative of a better ability to minimize false positives. However, this comes at the cost of recall, meaning it may miss many true hallucinations. Other models, like GaussianNB and DecisionTree, have much lower precisions of 9.83% and 18.52%, respectively, hence, are poor at avoiding false alarms in predicting hallucinations.

Recall, on the other hand, gives the model's ability to identify actual hallucinations. Here, AdaBoost does exceptionally well and has a recall of 60.07%, which may imply that it captures most true positive cases. However, this is offset by a much lower precision, leading to a high rate of false positives. This imbalance between precision and recall speaks volumes about the trade-offs which come with classifier optimization in this dataset. The F1-score, which harmonizes precision and recall, provides a clearer picture of overall predictive balance. While none of the models achieve a high F1-score, lgb.LGBM with 28.21% and AdaBoost with 23.38% strike a better balance between the competing demands of precision and recall compared to others. These results indicate that achieving both reliability and sensitivity remains a challenge in hallucination prediction. Additionally, the RoC_AUC represents the general discriminative power of a classifier in distinguishing between classes. The performances of CatBoost with 79.64%, and lgb.LGBM with 79.34%, has the largest overall discriminative powers, whereas DecisionTree turns out with the poorest result, only 59.02%.

Overall, the analysis underscores the importance of choosing classifiers based on the specific priorities of the application. For scenarios demanding high precision to avoid false positives, RandomForest stands out. Conversely, for applications requiring high recall to identify as many hallucinations as possible, AdaBoost is preferable. Models like lgb.LGBM and CatBoost, with their balanced performance across multiple metrics, offer promising options for general-purpose applications.

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Results from both the training and test datasets show a high-contrast landscape regarding the performance of the classifiers for hallucination prediction, with most models showing significant overfitting. While the performance metrics are almost perfect on the training dataset, they considerably degrade when applied to the test dataset. This disparity underlines challenges regarding generalization, robustness, and model complexity.

A. Cross validation Model

To improve the prediction models, a cross-validation approach is used instead of the split-test method. For brevity, two models—Random Forest and Decision Tree—are selected for enhancement and evaluation. The cross-validation method yielded high performance with the Random Forest classifier with an accuracy of 94.91%, a high precision of 84.34% and a moderate recall of 54.14%. These outcomes further confirm that the Random Forest model is accurate in terms of distinguishing positive cases since the f1 score is low because the model is unable to identify all actual positive scenarios. As for the Decision Tree classifier, the precision was 0.5880 while the recall was 0.5871 and the accuracy was 0.9163. The performance of these metrics indicates that the decision tree model exhibits similar performance over both the training and cross-validation but is slightly lower than the Random Forest model in the sense that it records slightly lower overall accuracy. Finally, the combining the models, the results revealed that:

- Cross-validation shows a reliable model in evaluating the hallucination cases, besides the model reduces the error from overfitting by validating on different subsets of the data.
- Due to the fixed test set, split-test validation can be skewed, thus reporting over-optimistic accuracy or underreporting of recall, especially for imbalanced datasets.

B. Results discussion and previous works

The comparison of our models with previous works on hallucination prediction has brought out both strengths and areas of improvement. Our models are quite robust on the test dataset, yielding an accuracy of 94.77% for RandomForest and 94.25% for Bagging, hence lying very close to the state-of-the-art results reported in studies like those of Liu et al. (2024) [12] and Kalavasis et al. (2024)[13]. As discussed in our research and based on the previous literature reviews, high accuracy does not reflect the full complexity of the hallucination detection problem, since imbalanced datasets are used.

A large divergence in precision and recall can be noticed. Our models showed a big trade-off: RandomForest has high precision (65%) but low recall (4.85%), reflecting a conservative prediction approach which minimizes false positives at the cost of missing true hallucinations. On the other hand, some of the approaches suggested by earlier works, such as retrieval-augmented generation or multimodal data grounding, aim to perform well across these metrics in a better manner. These often show more consistent scores of recall, at the cost of reduced precision, thus indicating the continued difficulty in the quest for superior trade-offs in this area [14].

The AUC scores of the ROC curves, 59-79% for our models, therefore denote their overall performance. These numbers remain on par with those presented in several research papers, one among which is that of Sriramanan et al. (2024) [15]. Our findings indicated that all the models retain the appropriate capacity for differentiating between the Hallucinated versus the Non-hallucinated ones. Methodologically, the work deploys a range of traditional classifiers like XGB, CatBoost, and AdaBoost, thus comparing the performance across different paradigms. In addition, most previous works have focused either on fine-tuning LLMs or using their ensembles to produce better results.

Finally, the collected results which compared with previous models in the field of hallucination detection, found that the proposed models did not follow the random guessing for the hallucination classes, and there is a strong indicator of overfitting in the trained data, which indicated that the proposed model cannot generalize well to unseen data. Therefore, to solve the weakness in the proposed model, many improvements can be done to solve that including:

1- Use an oversampling for hallucinations, and under-sampling for non-hallucinations.
2- Apply regularization techniques to reduce overfitting and improve generalization.
3- Use retrieval-augmented generation (RAG) or multimodal grounding to handle hallucination detection with imbalanced dataset.
4- Optimize the parameters of the training models to improve hallucination detection.
5- Reorganized the dataset to make it balanced by revising the labelled hallucination models.
6- Use different data splitting models such as cross-validation to improve the training models.

IV. Conclusion and Future Work

This work proposed a machine learning-based model for the detection of hallucinations in large language model systems. This study has adopted traditional machine learning techniques and found very promising results: the performance in terms of accuracy reached 94.77% on the test set by using the Random Forest algorithm. However, significant challenges persist regarding the balancing of performance across all metrics. A notable issue is that there is a significant difference between training and testing, and also the precision-recall trade-off points to overfitting. To improve the performance, cross-validation is used with random forest and decision tree, the results showed that the cross-validation

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is better than the test and train splitting model. In addition, the results of cross-validation can avoid overfitting.

Our key finding in this research can be summarized as:

- **Cross-validation:** the cross- validation method showed that it can reduce the overfitting cases.
- **Generalization Issues:** The model performance shows a big difference between training and testing, which means it is not generalizing well to unseen data.
- **Traditional ML Model Effectiveness**: The best results obtained using traditional machine learning methods are competitive with state-of-the-art methods for the detection of hallucinations.
- **Class Imbalance**: The class imbalance in distribution will critically affect the model's ability to maintain high performance across all metrics of evaluation.

The following represents some suggested directions for addressing these challenges:

- **Handling Class Imbalance**: Oversample and under-sample to reduce the adverse effects of an imbalanced dataset.
- **Reducing Overfitting**: Apply stronger regularization techniques and explore alternative data splitting strategies, such as cross-validation.
- **Advanced Techniques**: Understand the state-of-the-art methods of retrieval-augmented generation and multimodal grounding which lead to an improvement in the detection of hallucinations.
- **Hyperparameter Optimization**: Perform hyper parameter tuning to further improve model performance.
- **Improvement of Dataset Labeling**: Datasets should be reviewed for labelling methodologies so that all their classes are well-represented, and classes are better balanced.

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